

What is ERS

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What is ERS?

Electric Range Simulator (ERS) is a Volvo Group solution used to evaluate battery electric truck operations by estimating energy consumption and route feasibility. ERS combines vehicle data, route information and operational conditions to support transport planning and electrification decisions.

Purpose

ERS helps users understand whether a battery electric vehicle is suitable for a specific transport mission.

The solution supports:

- Vehicle configuration
- Route planning
- Travel plan management
- Energy consumption estimation
- Charging-related planning
- Simulation result analysis
- PDF report generation

ERS is used across multiple Volvo Group truck brands and supports different vehicle offerings and electrification scenarios.

What Problem Does ERS Solve?

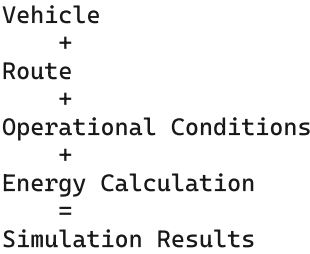
Transitioning from diesel to battery electric vehicles requires understanding:

- Expected energy consumption
- Route feasibility
- Vehicle suitability
- Charging requirements
- Operational constraints

ERS provides simulation capabilities that help evaluate these factors before real-world deployment.

How ERS Works

At a high level, ERS combines:



The platform uses vehicle configuration information, route data and energy calculation services to estimate expected energy consumption and operational outcomes.

Main Functional Areas

Area	Purpose
Vehicles	Configure and manage vehicle data
Garage	Manage reusable vehicle configurations
Route Planning	Create and manage routes and travel plans
Charging	Configure charging-related settings
Simulation	Estimate energy consumption and route feasibility
Reports	Generate and export simulation results

These capabilities are supported by core domains such as Vehicle Configuration, Route Management and Customer Management.

Core Building Blocks

Vehicle Configuration

Provides the vehicle information used during simulations.

Examples include:

- Vehicle Type
- Vehicle Model
- Battery Generation
- Trailer Configuration

Vehicle configuration is one of the core ERS domains.

Route Management

Route Management handles:

- Routes
- Travel Plans
- Waypoints
- Route inputs

Route data is used as input for energy calculations and simulation scenarios.

Travel Plans

A Travel Plan represents a simulation scenario that combines routes, vehicles and settings to evaluate a transportation mission.

Garage

Garage (Trucks in Garage / TiG) provides access to predefined vehicle configurations and vehicle-related data used within ERS workflows. Recent ERS roadmap discussions reference TiG as an important part of the future Energy API-based solution.

Key Integrations

ERS interacts with several external systems.

Integration	Purpose
Energy API	Energy consumption calculations
HERE Maps	Route and mapping capabilities
TiG	Vehicle and predefined truck information
PDF Export	Report generation and sharing
TIL API	Customer and dealership synchronization

These integrations are documented in ERS integration materials.

Energy API

Energy API is a strategic component of the ERS roadmap.

ERS is evolving toward stronger integration with Energy API for energy prediction and simulation capabilities. Program planning materials describe Energy API as the target calculation service for future ERS scenarios and offerings.

Who Uses ERS?

ERS supports users involved in:

- Customer transport analysis
- Electrification assessments
- Vehicle selection
- Route evaluation
- Sales support activities
- Product and engineering validation

ERS contributes to customer self-assessment and electrification-related decision support.

ERS Today and Future Direction

Current ERS capabilities continue to support vehicle simulation and planning, while future roadmap activities focus on deeper Energy API integration, TiG-based vehicle handling and support for future vehicle offerings. Program discussions describe a transition toward a new ERS concept that leverages Energy API-based calculations while maintaining access to existing functionality during the transition period.